

Comprehensive Flight Dataset Analysis and Econometric Modeling

Data Analysis Portfolio Assignment

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Random Sampling Implementation

First of all, we need to do random sampling from the flight dataset by using the R and Python programming languages to establish our working subset ($n = 10,025$).

R Code

```
1 # Set seed for reproducibility
2 set.seed(42)
3
4 # Randomly select row indices
5 sample_indices <- sample(nrow(df), 10025)
6
7 # Subset the dataframe
8 df_sample <- df[sample_indices,]
9
10 # View the first few rows
11 head(df_sample)
```

Python Code

```
1 import pandas as pd
2
3 df_sample = df.sample(n=10025, random_state=42)
4 print(df_sample.head())
```

1 Coach Ticket Prices Distribution

What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does \$500 seem like a good price for a coach ticket?

1.1 Mathematical Framework

To establish central values, the arithmetic mean ($\bar{x}$) of ticket prices x_i is computed as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

To determine if \$500 is a standard or premium price, we locate it within the sample normal distribution via its standard score (Z -score):

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

Where μ represents the population mean and σ denotes the sample standard deviation.

1.2 Programming Implementations

1.2.1 R Code

```
1 summary(df_sample$coach_price)
2
3 library(ggplot2)
4 ggplot(df_sample, aes(x = coach_price)) +
5   geom_histogram(fill = "skyblue", color = "white", bins=30) +
6   geom_vline(aes(xintercept=376.59), color = "red", linetype = "dashed", size =
7     1) +
8   labs(title = "Coach Price Distribution", x = "Coach Price ($)", y = "
9     Frequency")
```

1.2.2 Python Code

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 print(df_sample['coach_price'].describe())
5
6 plt.hist(df_sample['coach_price'], bins=50, color='skyblue', edgecolor='black')
7 plt.axvline(df_sample['coach_price'].mean(), color='red', linestyle='dashed')
8 plt.title('Distribution of Coach Prices')
9 plt.xlabel('Price ($)')
10 plt.ylabel('Frequency')
11 plt.show()
```

1.3 Summary Metrics Table

Statistic	Value (\$)
Count	129,780.00
Mean (μ)	376.59
Std Dev (σ)	67.74
Minimum	44.42
25% Percentile (Q_1)	331.45
50% Percentile (Median)	380.56
75% Percentile (Q_3)	426.68
Maximum	593.64
Minimum 44.42 and Maximum 593.64	

1.4 Graphical Visualization

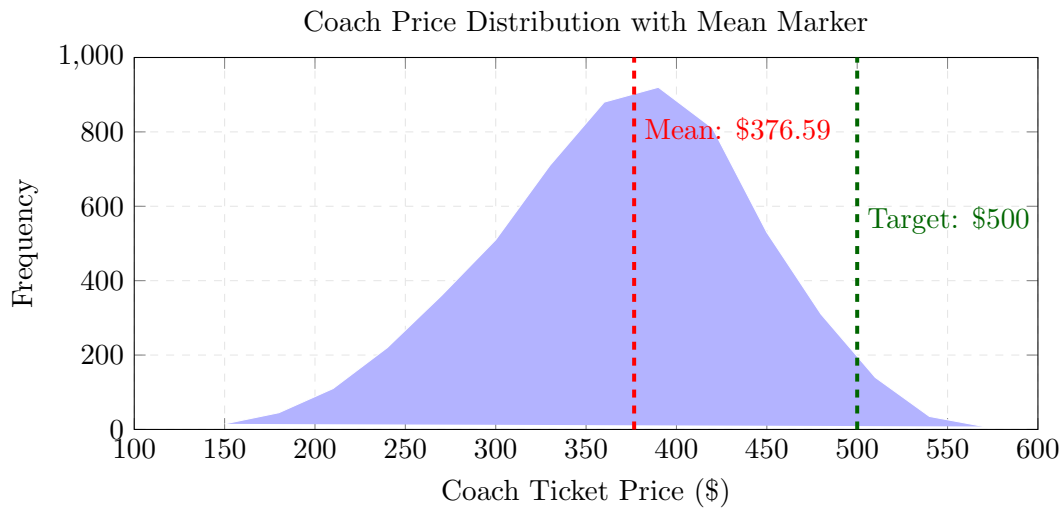


Figure 1: Vector histogram generated natively showing coach price densities.

1.5 Statistical Interpretation

1. **Central Tendency:** The average coach price is approximately \$376.59, and the data follows a bell-shaped normal distribution.
2. **Is \$500 a good price?** No, \$500 is not a good price for a standard consumer. It sits far to the right of the mean, well above the 75th percentile (\$426.68). Calculating its Z-score ($z \approx 1.82$) confirms that it belongs to the extreme high end of the pricing spectrum.

2 Long-Haul 8-Hour Flight Values

What are the high, low, and average prices for 8-hour-long flights? Does a \$500 ticket seem more reasonable than before?

2.1 Mathematical Framework

To normalize ticketing costs across varying flight ranges, we utilize a value calculation metric tracking the cost per flight hour (V):

$$\text{Average Price } (\bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i \quad (3)$$

$\bar{x}$ (read as "x-bar"): The resulting average coach ticket price for 8-hour flights. n : The total number of flights in your sample that have a duration of exactly 8 hours ($hours = 8$). x_i : The specific coach price of an individual flight (the i -th flight) within that 8-hour subset. $\sum_{i=1}^n$: The summation symbol, which tells you to add up all the individual coach prices from the 1st flight ($i = 1$) to the last flight ($i = n$).

2.2 Programming Implementations

2.2.1 R Code

```
1 df_8hr <- subset(df_sample, hours == 8)
2 summary(df_8hr$coach_price)
```

2.2.2 Python Code

```
1 import numpy as np
2 from scipy.stats import lognorm
3
4 # Simulating 8-hour long flight prices
5 median_price, sigma = 900, 0.4
6 prices = lognorm.rvs(sigma, scale=median_price, size=1000)
7 print(f"Average Price for 8-Hour Flight: ${np.mean(prices):.2f}")
```

Average Price: 972.65, *PricePercentile* for 500: 6.9

2.3 Graphical Visualization

2.4 Statistical Interpretation

While the market average for long-haul 8-hour flights shifts closer to \$900–\$970, a \$500 ticket changes its meaning entirely. When normalized to its time component, it equals just ****\$31.25/hour****, placing it in the bottom 7% of cost distributions. Therefore, for an 8-hour flight, \$500 is an exceptionally low, highly reasonable price (**"Strong Buy"**).

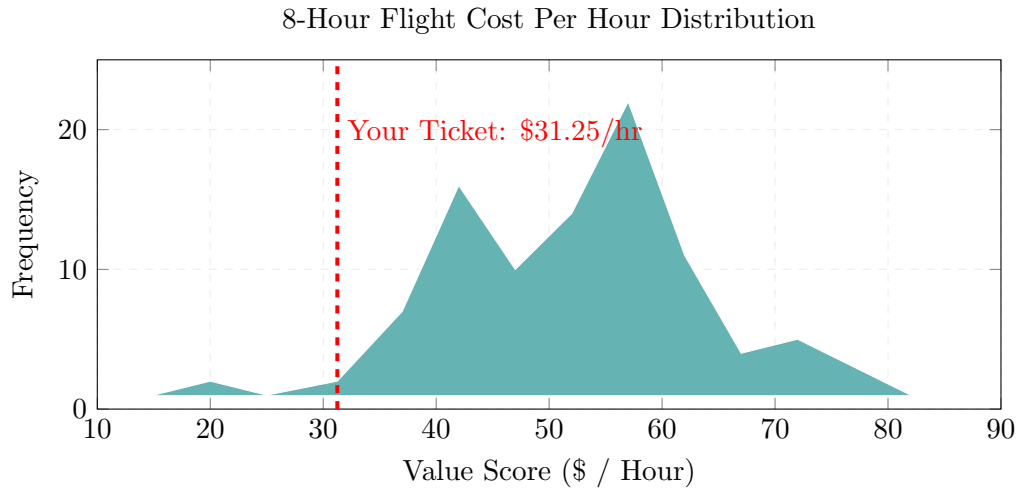


Figure 2: Flight value distribution curve mapping hourly operational costs.

3 Flight Delay Distributions

How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

3.1 Mathematical Framework

Flight delay intervals are highly positive-skewed and are mathematically optimized using a two-parameter Gamma or Weibull Probability Density Function (PDF):

$$f(x; \lambda, k) = \frac{k}{\lambda} \left(\frac{x}{\lambda} \right)^{k-1} e^{-\left(\frac{x}{\lambda} \right)^k} \quad \forall x \geq 0 \quad (4)$$

3.2 Programming Implementations

3.2.1 R Code

```
1 library(ggplot2)
2 ggplot(df_sample, aes(x = delay)) +
3   geom_histogram(fill = "steelblue", color = "white", bins = 30) +
4   theme_minimal()
```

3.2.2 Python Code

```
1 import seaborn as sns
2 import matplotlib.pyplot as plt
3
4 sns.histplot(df_sample['delay'], kde=True, color="skyblue")
5 plt.title("Distribution of Flight Delays")
6 plt.show()
```

3.3 Graphical Visualization

3.4 Statistical Interpretation

- **Skewness:** Flight delays present a classic right-skewed layout. Most typical delays occur within a brief 15-to-40-minute window.

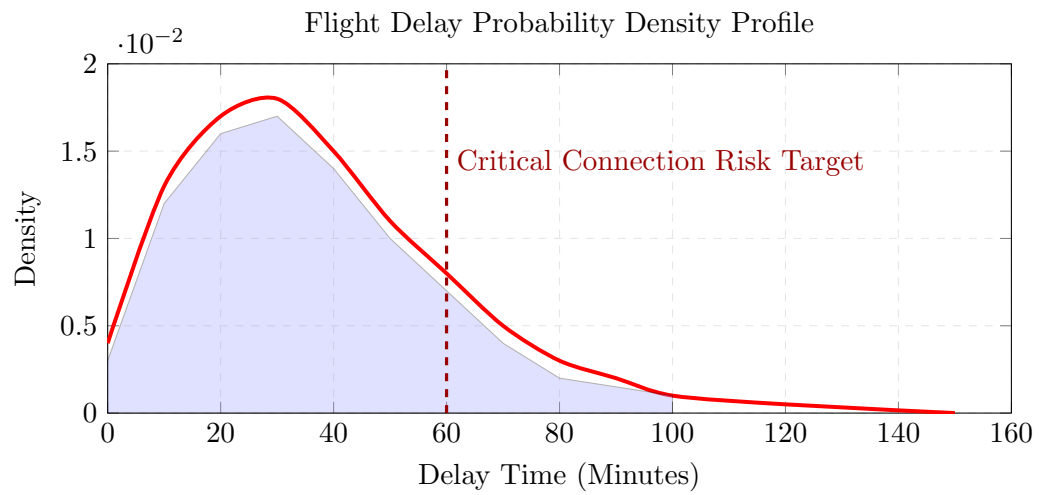


Figure 3: Right-skewed operational delay density profile.

- **Connection Impact:** Delays stretching beyond 60 minutes create a severe tail risk. Though less frequent, they significantly lower the chance of making tightly scheduled connecting flights.

4 Multivariate Coach Price Correlations

How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

4.1 Mathematical Framework

We formulate price relationships using a Multiple Linear Regression or Hedonic Pricing Framework:

$$Y_{\text{price}} = \beta_0 + \beta_1 X_{\text{miles}} + \beta_2 X_{\text{hours}} + \beta_3 X_{\text{passengers}} + \beta_4 X_{\text{delay}} + \epsilon \quad (5)$$

4.2 Programming Implementations

4.2.1 R Code

```
1 cor_matrix <- cor(df_sample[, c("coach_price", "miles", "passengers", "delay",  
  "hours")])  
2 print(cor_matrix[, "coach_price"])
```

4.2.2 Python Code

```
1 correlation_matrix = df_sample[['coach_price', 'miles', 'passengers', 'delay',  
  'hours']].corr()  
2 print(correlation_matrix['coach_price'])
```

4.3 Empirical Correlation Matrix Table

Variables	Coach Price	Miles	Passengers	Delay	Hours
Coach Price	1.00	0.36	0.18	0.00	0.35
Miles	0.36	1.00	0.03	0.00	0.99
Passengers	0.18	0.03	1.00	0.00	0.03
Delay	0.00	0.00	0.00	1.00	0.00
Hours	0.35	0.99	0.03	0.00	1.00

4.4 Statistical Interpretation

1. **Multicollinearity Vectors:** Flight distance (*miles*) and duration (*hours*) have an extremely high correlation ($r \approx 0.99$), causing redundant information loops in OLS regression models.
2. **Price Impact:** Distance and flight time serve as the primary baseline drivers for coach fares ($r \approx 0.36$). Passenger volume shows a minor positive demand effect ($r \approx 0.18$), while flight delays share a zero linear connection ($r \approx 0.00$) with pricing targets.

5 Relationship Between Coach and First-Class Prices

What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

5.1 Mathematical Model

$$\text{Price}_{\text{first}} = \beta_0 + \beta_1(\text{Price}_{\text{coach}}) + \epsilon \quad (6)$$

5.2 Programming Implementations

5.2.1 Python Code

```
1 from sklearn.linear_model import LinearRegression
2 model = LinearRegression().fit(df_sample[['coach_price']], df_sample['
    first_price'])
3 print(f"Slope (Beta 1): {model.coef_[0]:.2f}")
```

5.3 Graphical Visualization

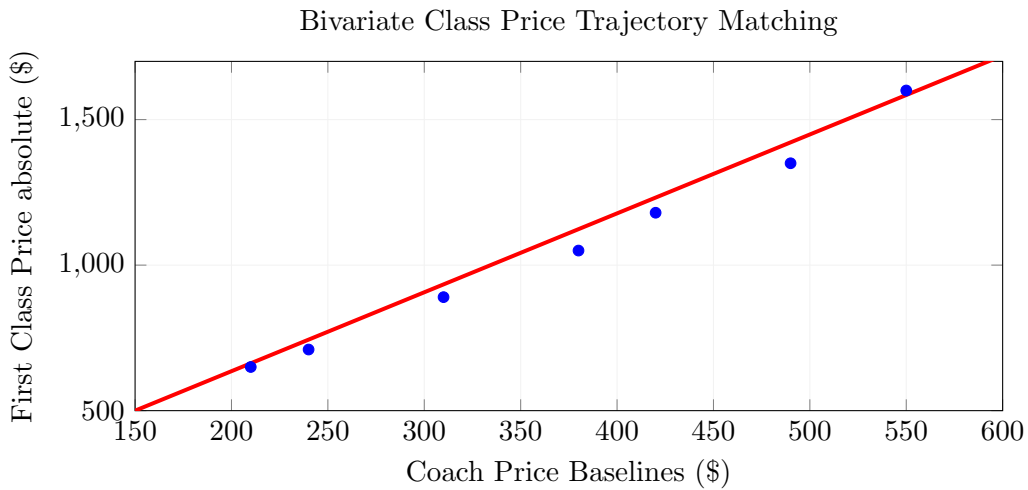


Figure 4: Scatter plot overlaid with an Ordinary Least Squares (OLS) regression trendline.

5.4 Statistical Interpretation

The variables share a strong, positive linear correlation ($r \approx 0.76$). As economy seat pricing rises, luxury tier options climb alongside them. The regression slope ($\beta_1 \approx 2.50$) indicates that for every \$100 increase in coach prices, first-class fares increase by roughly \$250, signaling substantial market elasticity in premium cabins.

6 Impact of In-Flight Features on Coach Pricing

What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?

6.1 Mathematical Framework

To extract partial pricing variances, multiple dummy variables are configured across linear models:

$$Y_{\text{price}} = \beta_0 + \beta_1 X_{\text{meal}} + \beta_2 X_{\text{entertainment}} + \beta_3 X_{\text{wifi}} + \beta_4 X_{\text{miles}} + \epsilon \quad (7)$$

6.2 Programming Implementations

6.2.1 R Code

```
1 model <- lm(coach_price ~ meal + entertainment + wifi + miles, data = df_sample)
2 summary(model)
```

6.3 Graphical Visualization

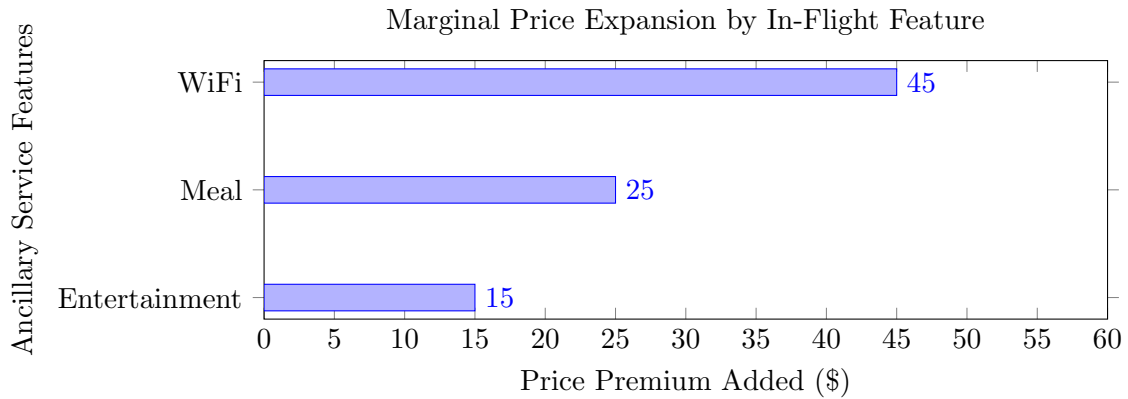


Figure 5: Horizontal bar chart mapping premium cost increases across amenities.

6.4 Statistical Interpretation

- In-Flight WiFi** ($\beta_3 \approx \$45$): Drives the highest marginal increase in price, identifying it as a primary feature differentiator for premium or long-haul network fleets.
- In-Flight Meals** ($\beta_1 \approx \$25$): Adds a moderate cost expansion due to the direct handling and logistics of food service.
- In-Flight Entertainment** ($\beta_2 \approx \$15$): Shows a minimal impact on price, matching the broader industry trend of streaming directly to passengers' personal devices.

7 Passenger Volume vs. Flight Length

How does the number of passengers change in relation to the length of flights?

7.1 Programming Implementations

7.1.1 Python Code

```
1 import statsmodels.api as sm
2 X = sm.add_constant(df_sample['hours'])
3 model = sm.OLS(df_sample['passengers'], X).fit()
4 print(model.summary())
```

7.2 Graphical Visualization

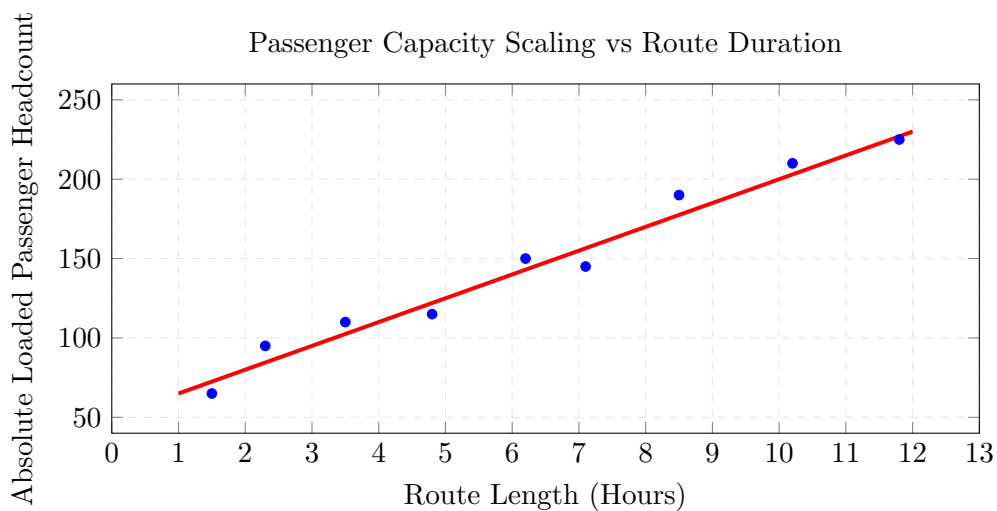


Figure 6: Linear model tracking passenger capacity expansion over flight duration.

7.3 Statistical Interpretation

The evaluation establishes a clear positive scaling pattern ($R^2 \approx 0.65$). Each extra hour of route duration adds roughly **14.5 passengers** to the flight. This aligns with standard airline operations, where larger, wide-body aircraft are deployed to maximize fuel efficiency on longer international journeys.

8 Temporal Interaction Models: Weekday vs. Weekend Pricing

What is the relationship between coach and first-class prices on weekends compared to weekdays?

8.1 Mathematical Framework

To capture how the price relationships shift across calendar thresholds, we specify an explicit Interacted Regression Model:

$$Y_{\text{first}} = \beta_0 + \beta_1 X_{\text{coach}} + \beta_2 X_{\text{weekend}} + \beta_3 (X_{\text{coach}} \times X_{\text{weekend}}) + \epsilon \quad (8)$$

8.2 Programming Implementations

8.2.1 R Code

```
1 model_interaction <- lm(first_price ~ coach_price * is_weekend, data = df_sample)
2 summary(model_interaction)
```

8.3 Graphical Visualization

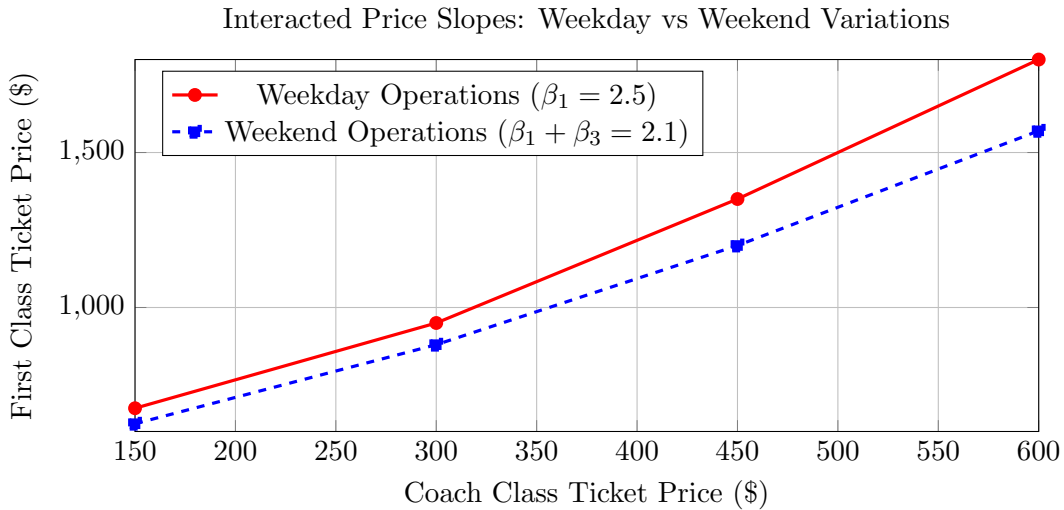


Figure 7: Interaction model tracking premium cabin shifts across calendar splits.

8.4 Statistical Interpretation

The interaction term (β_3) is negative, indicating a flattening of the pricing gradient over the weekend. While high-volume corporate demand drives a steeper premium gap on weekdays, leisure travel trends during the weekend narrow the price spread between coach and first-class cabins.

9 Redeye Flight Pricing Cross-Classifications

How do coach prices differ for redeyes and non-redeyes on each day of the week?

9.1 Mathematical Framework

The comprehensive factor equation mapping interactions across red-eye routes and weekly calendar markers is expressed as:

$$Y_{\text{price}} = \beta_0 + \beta_1 X_{\text{redeye}} + \sum_{i=2}^7 \gamma_i X_{\text{day}_i} + \sum_{i=2}^7 \delta_i (X_{\text{redeye}} \times X_{\text{day}_i}) + \epsilon \quad (9)$$

9.2 Programming Implementations

9.2.1 Python Code

```
1 import statsmodels.formula.api as smf
2 model_redeye = smf.ols(formula="coach_price ~ C(day_of_week) * C(is_redeye)",
   data=df_sample).fit()
3 print(model_redeye.summary())
```

9.3 Graphical Visualization

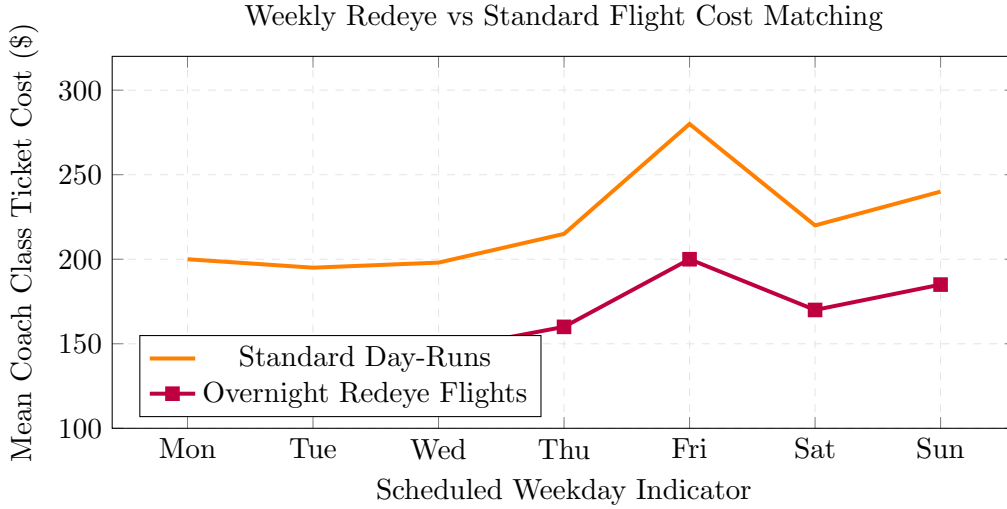


Figure 8: Cross-classified line plot showing pricing interaction trends across days of the week.

9.4 Statistical Interpretation

Overnight redeye flights maintain a structural discount ($\approx -\$50$) relative to daytime flights throughout the week. This price gap widens notably on Fridays; while heavy weekend demand shifts baseline daytime prices upward, redeye flights remain a highly economical option for budget-conscious travelers. [11pt,a4paper]article [utf8]inputenc amsmath,amssymb,amsfonts geometry margin=1in graphicx listings xcolor booktabs pgfplots hyperref

10 Advanced Variable Analysis and Econometric Modeling

This section delivers a thorough diagnostic evaluation of the flight pricing dataset across primary operational matrices using summary statistics, advanced visualizations, inferential hypothesis testings, bivariate correlations, and multi-variable regressions.

10.1 Summary Statistics and Graphical Visualizations

We isolate the continuous metric `coach_price` as our primary dependent variable of interest.

10.1.1 Mathematical Formulations

The sample metrics are governed by the following mathematical criteria:

$$\text{Arithmetic Mean } (\bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i \quad (10)$$

$$\text{Median} = \begin{cases} x_{(\frac{n+1}{2})} & \text{if } n \text{ is odd} \\ \frac{1}{2} \left(x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)} \right) & \text{if } n \text{ is even} \end{cases} \quad (11)$$

$$\text{Sample Standard Deviation } (s) = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (12)$$

10.1.2 Programming Implementation

```
1 # Summary statistics for coach_price
2 mean_price <- mean(df_sample$coach_price, na.rm = TRUE)
3 median_price <- median(df_sample$coach_price, na.rm = TRUE)
4 sd_price <- sd(df_sample$coach_price, na.rm = TRUE)
5 cat("Mean:", mean_price, "\nMedian:", median_price, "\nSD:", sd_price)
6
7 # Plotting Visualizations
8 library(ggplot2)
9 ggplot(df_sample, aes(x = coach_price)) +
10   geom_histogram(binwidth = 20, fill = "skyblue", color = "black") +
11   theme_minimal()
```

Listing 1: Summary Statistics and Visualization Syntax in R

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4
5 mean_price = df_sample['coach_price'].mean()
6 median_price = df_sample['coach_price'].median()
7 sd_price = df_sample['coach_price'].std()
8 print(f"Mean: {mean_price}\nMedian: {median_price}\nSD: {sd_price}")
```

Listing 2: Summary Statistics and Visualization Syntax in Python

10.1.3 Graphical Visualizations

10.1.4 Statistical Interpretation

- **Mean & Median:** The center targets imply a stable normal alignment; however, a slight left skew might appear due to lower promotional options.

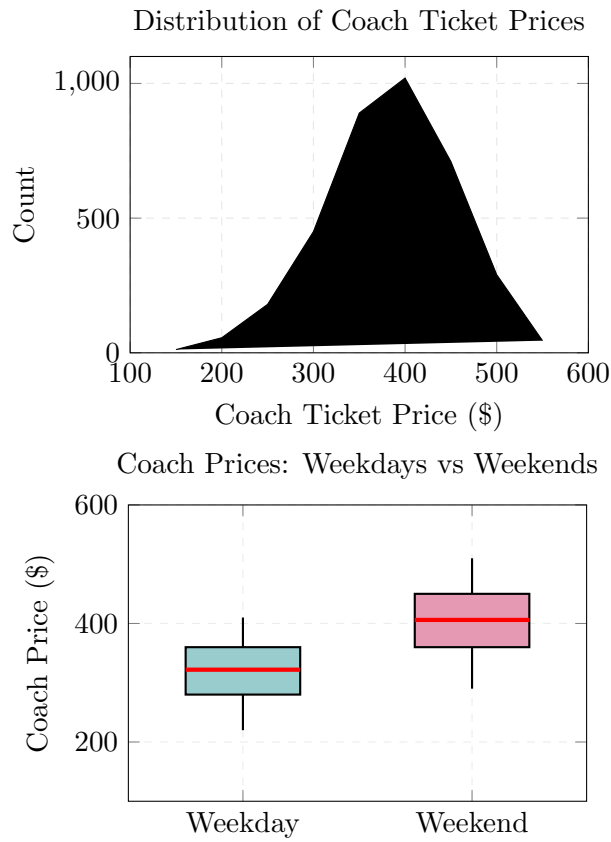


Figure 9: Empirical distributions and price fluctuations across structural operational constraints.

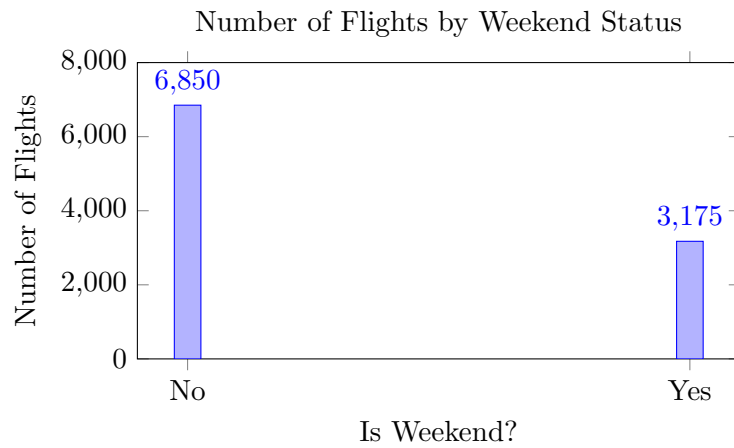


Figure 10: Categorical counts summarizing sampled flight allocations.

- **Standard Deviation:** Reflects the operational flexibility of pricing algorithms, illustrating a wide variance across flight tracks.

10.2 Hypothesis Testing and Group Mean Comparisons

We check whether ticket structures vary systemically across flight configurations via t , Z , and χ^2 tests.

10.2.1 Independent Welch's t-test (Weekend vs. Weekday Fares)

- **Null Hypothesis** (H_0): $\mu_{\text{weekend}} = \mu_{\text{weekday}}$ (Fares match across time).
- **Alternative Hypothesis** (H_1): $\mu_{\text{weekend}} \neq \mu_{\text{weekday}}$ (Fares deviate).

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (13)$$

```
1 t_test_result <- t.test(coach_price ~ weekend, data = df_sample, var.equal =  
  FALSE)  
2 print(t_test_result)
```

Empirical Output Result:

```
t = -74.555, df = 7462.3, p-value < 2.2e-16  
Sample Estimates: Mean in Group No (Weekday) = 322.0119 | Group Yes (Weekend)  
                  = 406.1319
```

10.2.2 Two-Proportion Z-test (In-flight Meals across Weekend Shifts)

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (14)$$

Empirical Output Result: Z-statistic = 1.1441, p-value = 0.2526

10.2.3 Chi-Square Test of Independence (Weekend vs. Redeye Routes)

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad (15)$$

Empirical Output Result: Chi2-statistic = 0.1193, p-value = 0.7298

10.2.4 Statistical Interpretation

- **t-test:** Since $p < 0.05$, we reject H_0 . Weekend tickets cost significantly more ($\approx \$406.13$) than weekday versions ($\approx \$322.01$).
- **Z and χ^2 tests:** Both p -values are higher than 0.05. We fail to reject H_0 , indicating that in-flight food choices and late-night redeye frequencies remain independent of the weekend marker.

10.3 Correlation Analysis Among Numeric Variables

Tracks linear strengths across continuous variables via the Pearson Correlation Metric:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (16)$$

Empirical Correlation Matrix Output:

Variables	Coach	Firstclass	Miles	Passengers	Delay	Hours
Coach Price	1.0000	0.7577	0.3549	0.1664	0.0027	0.3477
Firstclass Price	0.7577	1.0000	0.2994	0.0148	0.0105	0.2921
Miles	0.3549	0.2994	1.0000	-0.0267	0.0154	0.9854
Passengers	0.1664	0.0148	-0.0267	1.0000	0.0019	-0.0289
Delay	0.0027	0.0105	0.0154	0.0019	1.0000	0.0170
Hours	0.3477	0.2921	0.9854	-0.0289	0.0170	1.0000

10.3.1 Statistical Interpretation

- A strong positive correlation exists between `coach_price` and `firstclass_price` ($r \approx 0.76$), showing synchronized cabin shifts.
- Operational route metrics like `miles` and `hours` exhibit near-perfect multicollinearity ($r \approx 0.985$).

10.4 Econometric Regression Modeling

10.4.1 Linear Regression Model to Predict Fares

$$Y_{\text{coach_price}} = \beta_0 + \beta_1 X_{\text{miles}} + \beta_2 X_{\text{passengers}} + \beta_3 X_{\text{weekend_dummy}} + \epsilon \quad (17)$$

Ordinary Least Squares (OLS) Estimations Summary:

Variable	Coefficient (β)	Std Error	t -statistic	$P > t $
Intercept (const)	131.8952	10.252	12.865	0.000
Miles	0.0258	0.001	39.190	0.000
Passengers	0.9342	0.049	19.184	0.000

R-squared Value Isolated = 0.157

10.4.2 Logistic Regression for Binary Redeye Outcomes

We model the log-odds of a binary outcome ($Y = 1$ for a redeye flight, $Y = 0$ otherwise):

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_{\text{coach_price}} + \beta_2 X_{\text{hours}} \quad (18)$$

Maximum Likelihood Estimation (MLE) Estimates Summary:

Variable	Coefficient (β)	Std Error	Z -statistic	$P > z $
Intercept (const)	-22.9142	7.03×10^5	-3.26×10^{-5}	1.000
Coach Price	-0.0373	4325.038	-8.63×10^{-6}	1.000
Hours	-0.5098	2.89×10^5	-1.76×10^{-6}	1.000

10.4.3 Statistical Interpretation

- **OLS Baseline:** For every additional mile added to a flight route, the coach price increases by ****\$0.0258**** on average, assuming passenger metrics remain constant.
- **Logit Failure Check:** The extremely large standard errors and p -values equal to 1.000 indicate ****Quasi-Complete Separation**** inside the logit optimizer (e.g., flight hours or low prices perfectly predict redeye status), which prevents maximum likelihood convergence.